

JONATHAN JONGBIN KIM

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EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

B.S. in Electrical Engineering

Daejeon, Republic of Korea

Mar. 2025 – Present

GPA: 3.65 / 4.3 | **Double Major:** Computer Science (in progress)

Relevant Coursework (completed and in progress): Introduction to Reinforcement Learning, Introduction to Generative Models, Machine Learning Basics and Practices, Probability and Introductory Random Processes, Programming Languages, Analysis I, Linear Algebra, Data Structures and Algorithms, Signals and Systems

RESEARCH INTERESTS

Foundational Machine Learning; World Models and Model-Based Reinforcement Learning; Learning Theory; Theoretical Computer Science; Mathematics.

Exploring learning, reasoning, and computation through generalization, abstraction, memory, and planning.

SELECTED PROJECTS

3D Gaussian Splatting — Scene Reconstruction [GitHub]

2026

PyTorch renderer implementation and gsplat-based 3D reconstruction

- Implemented a pure-PyTorch 3DGS renderer with 3D covariance, perspective projection, depth sorting, and differentiable alpha compositing.
- Built a COLMAP-to-gsplat training pipeline with SfM initialization, L1/SSIM losses, progressive spherical harmonics, and densification hooks.
- Added ablation controls for densification, parameter freezing, and scale/opacity regularization, with TensorBoard logging and PSNR/L1 validation.
- Trained on 135 registered views for 30,000 steps on a Tesla T4, achieving validation PSNR of 27.354 dB and L1 error of 0.03535 with 505,695 Gaussians.

StarGAN v2 — Multi-Domain Image Translation [GitHub]

2025

Paper-based implementation and empirical evaluation

- Implemented a PyTorch framework for 256×256 multi-domain image translation on AFHQ, including a generator, mapping network, style encoder, and discriminator.
- Combined AdaIN-based style conditioning with adversarial, style reconstruction, diversity-sensitive, and cycle consistency losses in the training pipeline.
- Trained and evaluated the model on a MacBook Pro with an Apple M4 Pro and 24 GB RAM using the PyTorch MPS backend.
- Evaluated a 20,000-step checkpoint on 3,000 generated images, achieving an overall FID of 44.11 and LPIPS diversity of 0.4270; analyzed domain-specific generation quality and visual changes across training checkpoints.

SELECTED PUBLICATIONS

[1] **J. Kim**, et al., “Analyzing The Impact and Evaluation of Flexitime on Subway Congestion Using SARIMA Model”, *Proceedings of the Korea Computer Congress (KCC)*, 2024.

[2] **J. Kim**, et al., “Comparison of Deep Learning Architecture Performance and Suggestion of Baseline Model for Diagnosing Cassava Leaf Disease”, *Proceedings of the Korea Software Congress (KSC)*, 2024.

HONORS & AWARDS

Excellence Award, Junior Division

2024

Undergraduate/Junior Paper Competition, Korea Software Congress (KSC)

SKILLS

Programming: Python, C++, Bash

Machine Learning: PyTorch, NumPy

Tools: Git, Linux, CUDA, L^AT_EX